



VIMANSH MAHAJAN

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[LinkedIn](#) | [Portfolio](#) | [Github](#)



EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi)	CGPA: 9.28 / 10.00	2022 - Ongoing
<i>Bachelor of Technology (B.Tech.) in Computer Science Engineering (CSE)</i> <i>Achievements:</i> IIITD Dean's List – Academics (2024–25); National Semi-Finalist, Flipkart GRiD 7.0 (2024); Top 5% Finalist, Adobe GenSolve Hackathon (2024); <i>Organiser</i>, TEDxIIITD; <i>Member</i>, Placement Cell; <i>Batch Representative</i>, Student Senate		
The Vivekanand School, Narela, New Delhi	CBSE XII: 97.2%	2022
<i>Achievements:</i> JEE Mains 2022 – 99%ile (AIR 9,200); JEE Advanced 2022 – Qualified (AIR 10,109)		
Delhi Public School, Noida	CBSE X: 96.8%	2020

PROFESSIONAL EXPERIENCE

Bank of New York (BNY) Software Engineer Intern	May 2025- July 2025
- Built and deployed a content automation platform using Spring Boot, Python-integrated LLM pipeline, and Oracle DB, with Angular frontend transforming unstructured finance notes (text/images) into publish-ready articles. - Implemented domain-aware segmentation and contextual visual placement; containerized and deployed services on Google App Engine.	
Complex Systems (CoSy) Lab UG Researcher	Jan 2025- May 2025
- Developed RecipeGPT using Recipe1M dataset, achieving BERTScore > 0.85 and ROUGE score of 0.32. - Applied a custom NER model to extract structured ingredient entities, enhancing input consistency and model understanding. - Developed a 3-stage GPT-2 fine-tuning pipeline: pretrained on 100k+ recipe instructions, conditioned on raw ingredients and titles, then fine-tuned with NER-enhanced inputs using special tokens and label masking.	
TavLab UG Researcher	Aug 2024- Dec 2024
- Predicted IC50 values for HDAC6 inhibitors using ADMET and BlueDesc datasets with molecular descriptor-based modelling. - Built and optimized ML models with advanced preprocessing, feature selection, and statistical analysis to accelerate compound screening and drug candidate identification.	

SKILLS

Tools & Technologies: Python, Hugging Face Transformers, Java, Spring Boot, C++, PostgreSQL, OracleDB, MySQL, REST APIs, Git

Technical Skills: Machine Learning (ML), Artificial Intelligence (AI), Natural Language Processing (NLP), Data Science, Data Structures & Algorithms, Object-Oriented Programming, Database Management System (DBMS), System Design, Computer Networks

Certification: [Quantitative Finance & Algorithmic Trading in Python](#), [Ethical Hacking](#), [The Complete Python Bootcamp \(Udemy\)](#)

Soft Skills: Teamwork, Effective Communication, Adaptability & Rapid Learning, Problem Solving, Execution Under Pressure

PROJECTS

<u>Healthcare Answer Summarisation</u> <i>Python, PyTorch, Hugging Face Transformers, LoRA</i>	Apr 2025
- Built a BART-large-CNN + LoRA pipeline to generate perspective-specific summaries (Information, Suggestion, Experience, Question, Cause) from medical Q&A data. - Designed a dual-head classifier and improved generalisation by selectively fine-tuning on hard examples with an initial BERTScore < 0.84. - Outperformed a FlanT5 baseline by +3.24 BLEU and +0.0355 BERTScore (final: BLEU 5.61, BERTScore 0.8782).	
<u>OptiWealth: Quantitative Portfolio Optimization Platform</u> <i>Spring Boot, Python, yfinance, React, PostgreSQL</i>	Nov 2025
Designed and implemented a distributed portfolio analytics system with Spring Boot backend orchestrating Python microservices, computing multi-portfolio metrics: returns, volatility, Sharpe ratio, VaR/CVaR, beta, max drawdown, and diversification. Implemented time-series forecasting and risk modelling using ARIMA (price trends), GARCH (volatility), and Monte Carlo simulations, enabling forward-looking scenario analysis. Developed portfolio optimization pipelines (efficient frontier simulation, max-Sharpe, min-volatility) with scheduled market analysis and AI-generated investment summaries.	
<u>OncoHelp-AI: Medical Decision Support for Brain MRI</u> <i>Python, Streamlit, Grad-CAM, LLMs</i>	Dec 2025
- Built an AI-assisted brain MRI triage system using ResNet18 and EfficientNet models with Grad-CAM explainability and uncertainty estimation via Monte Carlo Dropout. - Developed a safe, context-aware medical summarization and Q&A layer using Gemini/BioGPT LLM, enforcing validated inputs and scoped responses for responsible AI usage.	

EXTRACURRICULARS

- Coding Platforms:** [Leetcode](#), [Codeforces](#) (*Pupil- Max Rating: 1275*)
- Hobbies:** Debate, Fitness, Badminton
- Volunteering Experience:** Volunteer at Seva Sankalp NGO